MACHINE LEARNING LAB MANUAL

AMRITA
VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Department of Computer Science and Engineering

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Aim : To perform exploratory data analysis (EDA) on a student performance dataset using Pandas and Matplotlib.

Dataset : Create a CSV file named student_data.csv with the following data:

RollNo	Name	Gender	Attendance	InternalMarks	ExternalMarks
101	Ravi	M	85	32	55
102	Priya	F	92	38	58
103	Kiran	M	78	28	45
104	Anu	F	95	40	60
105	Sita	F	88	35	54
106	Raj	M	70	25	40
107	Divya	F	90	37	57
108	Arun	M	82	30	48
109	Keerthi	F	96	39	59
110	Naveen	M	75	27	44

Tasks

Task A: Dataset Loading and Inspection

- Import the necessary Python libraries.
- Load the dataset into a Pandas DataFrame.
- Display the first five records.
- Display the last five records.
- Find the total number of rows and columns.
- Display the names of all columns.
- Display the data type of each column.
- Generate dataset information using info().

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("student_data.csv")

print(df.head())
```

print(df.tail())

print(df.shape)

print(df.columns)

print (df.dtypes)

print (df.info())

Output:

```
   RollNo   Name Gender  Attendance  InternalMarks  ExternalMarks
0     101   Ravi      M          85             32             55
1     102  Priya      F          92             38             58
2     103  Kiran      M          78             28             45
3     104    Anu      F          95             40             60
4     105   Sita      F          88             35             54
```

```
   RollNo     Name Gender  Attendance  InternalMarks  ExternalMarks
5     106      Raj      M          70             25             40
6     107    Divya      F          90             37             57
7     108     Arun      M          82             30             48
8     109  Keerthi      F          96             39             59
9     110   Naveen      M          75             27             44
```

```
(10, 6)

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```

```
Index(['RollNo', 'Name', 'Gender', 'Attendance', 'InternalMarks',
       'ExternalMarks'],
      dtype='str')
85.1
33.1
```

```
RollNo            int64
Name                str
Gender              str
Attendance        int64
InternalMarks     int64
ExternalMarks     int64
dtype: object
85.1
33.1
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   RollNo         10 non-null     int64
 1   Name           10 non-null     str
 2   Gender         10 non-null     str
 3   Attendance     10 non-null     int64
 4   InternalMarks  10 non-null     int64
 5   ExternalMarks  10 non-null     int64
dtypes: int64(4), str(2)
memory usage: 612.0 bytes
None
85.1
33.1

[Done] exited with code=0 in 1.426 seconds
```

Task B: Statistical Analysis

- Calculate the mean attendance of all students.
- Calculate the average Internal Marks.
- Calculate the average External Marks.
- Find the maximum Internal Marks.

- Find the minimum Internal Marks.
- Find the maximum External Marks.
- Find the minimum External Marks.
- Calculate the standard deviation of Internal Marks.
- Generate summary statistics using describe().

Code:

```
print(df["Attendance"].mean())
print(df["InternalMarks"].mean())
print(df["ExternalMarks"].mean())
print(df["InternalMarks"].max())
print(df["InternalMarks"].min())
print(df["ExternalMarks"].max())
print(df["ExternalMarks"].min())
print(df["ExternalMarks"].std())
print(df["InternalMarks"].std())
print(df.describe())
```

Output:

```
85.1
33.1
52.0
40
25
60
40
7.1492035298424055
5.425249610233001
           RollNo  Attendance  InternalMarks  ExternalMarks
count   10.00000   10.000000       10.00000      10.000000
mean   105.50000   85.100000       33.10000      52.000000
std      3.02765    8.736259        5.42525       7.149204
min    101.00000   70.000000       25.00000      40.000000
25%    103.25000   79.000000       28.50000      45.750000
50%    105.50000   86.500000       33.50000      54.500000
75%    107.75000   91.500000       37.75000      57.750000
max    110.00000   96.000000       40.00000      60.000000
```

Task C: Data Manipulation

- Create a new column named TotalMarks.
- TotalMarks=InternalMarks+ExternalMarks
- Create a new column named Percentage.

1. Percentage=TotalMarks/100×100 • Display the updated DataFrame.

Code:

df["TotalMarks"] = df["InternalMarks"] + df["ExternalMarks"]

df["Percentage"] = (df["TotalMarks"] / 100) * 100

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

[Running] python -u "c:\Users\Sreedurga\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\ML\task1.py"
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
0     101     Ravi      M  ...             55          87        87.0
1     102    Priya      F  ...             58          96        96.0
2     103    Kiran      M  ...             45          73        73.0
3     104      Anu      F  ...             60         100       100.0
4     105     Sita      F  ...             54          89        89.0
5     106      Raj      M  ...             40          65        65.0
6     107    Divya      F  ...             57          94        94.0
7     108     Arun      M  ...             48          78        78.0
8     109  Keerthi      F  ...             59          98        98.0
9     110   Naveen      M  ...             44          71        71.0

[10 rows x 8 columns]
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
1     102    Priya      F  ...             58          96        96.0
3     104      Anu      F  ...             60         100       100.0
8     109  Keerthi      F  ...             59          98        98.0
```

Task D: Filtering and Searching

- Display the details of students whose attendance is greater than 90%.
- Display the details of students whose TotalMarks exceed 90.
- Display the details of female students.
- Display the details of male students.
- Display students whose ExternalMarks are less than 50.

Code:

print(df[df["Attendance"]>90])

print(df[df["TotalMarks"]>90])

print(df[df["Gender"] == "F"])

print(df[df["Male"] == "M"])

print(df[df["ExternalMarks"]<50])

Output:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
[10 rows x 8 columns]
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
1     102    Priya      F  ...             58          96        96.0
3     104      Anu      F  ...             60         100       100.0
8     109  Keerthi      F  ...             59          98        98.0

[3 rows x 8 columns]
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
1     102    Priya      F  ...             58          96        96.0
3     104      Anu      F  ...             60         100       100.0
6     107    Divya      F  ...             57          94        94.0
8     109  Keerthi      F  ...             59          98        98.0

[4 rows x 8 columns]
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
1     102    Priya      F  ...             58          96        96.0
3     104      Anu      F  ...             60         100       100.0
4     105     Sita      F  ...             54          89        89.0
6     107    Divya      F  ...             57          94        94.0
8     109  Keerthi      F  ...             59          98        98.0

[5 rows x 8 columns]
   RollNo    Name Gender  ...  ExternalMarks  TotalMarks  Percentage
0     101    Ravi      M  ...             55          87        87.0
2     103   Kiran      M  ...             45          73        73.0
5     106     Raj      M  ...             40          65        65.0
7     108    Arun      M  ...             48          78        78.0
9     110  Naveen      M  ...             44          71        71.0

[5 rows x 8 columns]
   RollNo    Name Gender  ...  ExternalMarks  TotalMarks  Percentage
2     103   Kiran      M  ...             45          73        73.0
5     106     Raj      M  ...             40          65        65.0
7     108    Arun      M  ...             48          78        78.0
9     110  Naveen      M  ...             44          71        71.0
```

Task E: Ranking and Comparison

- Identify the student with the highest TotalMarks.
- Identify the student with the lowest TotalMarks.
- Display the top three performers based on TotalMarks.
- Display the bottom three performers based on TotalMarks.
- Determine how many students scored above the class average.

Code:

print(df.loc[df["TotalMarks"].idxmax()])

print(df.loc[df["TotalMarks"].idxmin()])

print(df.nlargest(3, "TotalMarks"))

print(df.nsmallest(3, "TotalMarks"))

average = df["TotalMarks"].mean()

print("\nStudents Above Class Average")

print(df[df["TotalMarks"] > average])

print("Number of Students:", len(df[df["TotalMarks"] > average]))

Output:

```
RollNo            104
Name              Anu
Gender              F
Attendance         95
InternalMarks      40
ExternalMarks      60
TotalMarks        100
Percentage      100.0
Name: 3, dtype: object
RollNo           106
Name             Raj
Gender             M
Attendance        70
InternalMarks     25
ExternalMarks     40
TotalMarks        65
Percentage      65.0
Name: 5, dtype: object
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
3     104      Anu      F  ...             60         100       100.0
8     109  Keerthi      F  ...             59          98        98.0
1     102    Priya      F  ...             58          96        96.0

[3 rows x 8 columns]
   RollNo    Name Gender  ...  ExternalMarks  TotalMarks  Percentage
5     106     Raj      M  ...             40          65        65.0
9     110  Naveen      M  ...             44          71        71.0
2     103   Kiran      M  ...             45          73        73.0

[3 rows x 8 columns]

Students Above Class Average
   RollNo     Name Gender  ...  ExternalMarks  TotalMarks  Percentage
0     101     Ravi      M  ...             55          87        87.0
1     102    Priya      F  ...             58          96        96.0
3     104      Anu      F  ...             60         100       100.0
4     105     Sita      F  ...             54          89        89.0
6     107    Divya      F  ...             57          94        94.0
8     109  Keerthi      F  ...             59          98        98.0

[6 rows x 8 columns]
Number of Students: 6

[Done] exited with code=0 in 1.599 seconds
```

Task F: Group-wise Analysis

- Calculate the average TotalMarks of male students.
- Calculate the average TotalMarks of female students.
- Compare the average attendance of male and female students.

Code:

```
print(df[df["Gender"] == "M"]["TotalMarks"].mean())

print(df[df["Gender"] == "F"]["TotalMarks"].mean())

print(df.groupby("Gender")["Attendance"].mean())
```

Output:

```
74.8
95.4
Gender
F    92.2
M    78.0
Name: Attendance, dtype: float64

[Done] exited with code=0 in 1.487 seconds
```

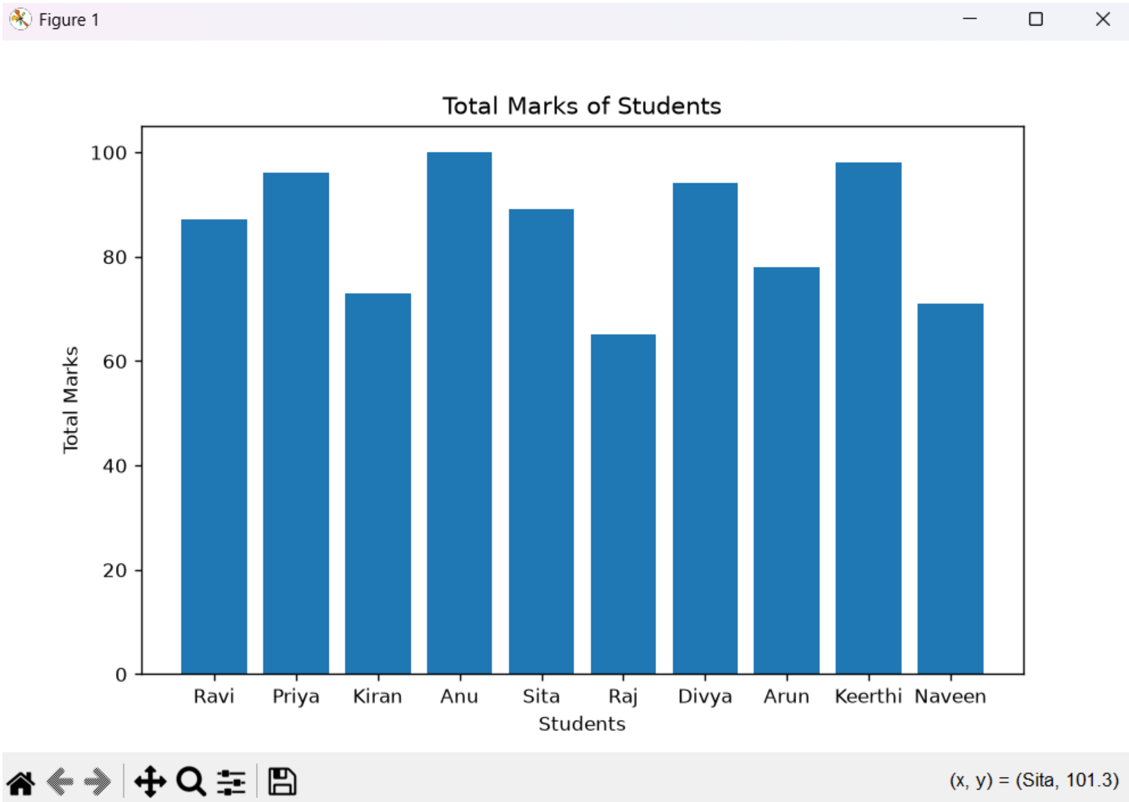
Task G: Data Visualization

- Plot a bar chart showing TotalMarks of all students.
- Plot a histogram of Attendance.
- Plot a pie chart showing the distribution of male and female students.
- Plot a scatter plot between Attendance and TotalMarks.
- Interpret the relationship between Attendance and TotalMarks.

Bar Chart Code:

```
plt.figure(figsize=(8,5))

plt.bar(df["Name"], df["TotalMarks"])

plt.title("Total Marks of Students")

plt.xlabel("Students")

plt.ylabel("Total Marks")

plt.show()
```

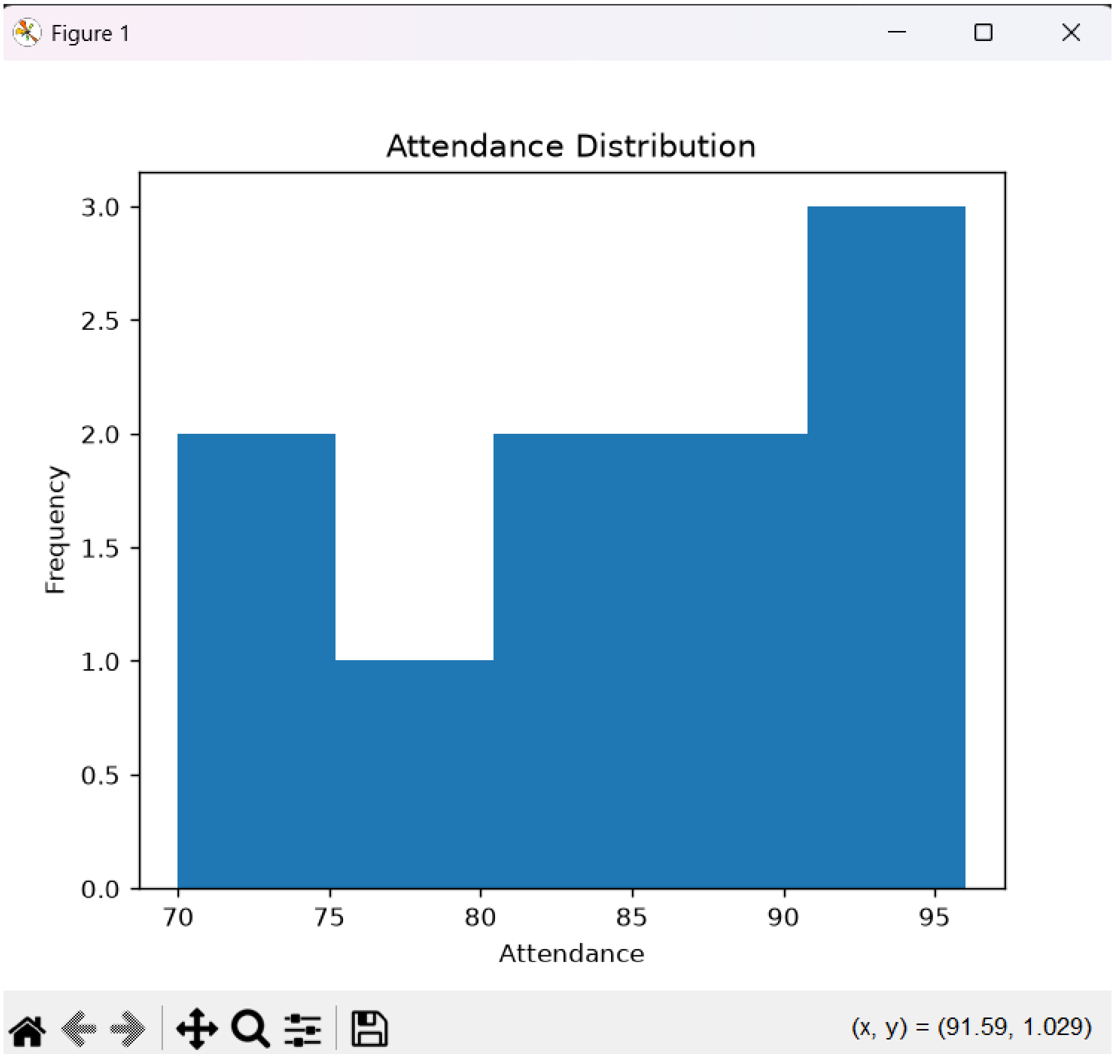
Output:



Histogram Code:

```
plt.figure(figsize=(6,5))

plt.hist(df["Attendance"], bins=5)

plt.title("Attendance Distribution")

plt.xlabel("Attendance")

plt.ylabel("Frequency")

plt.show()
```

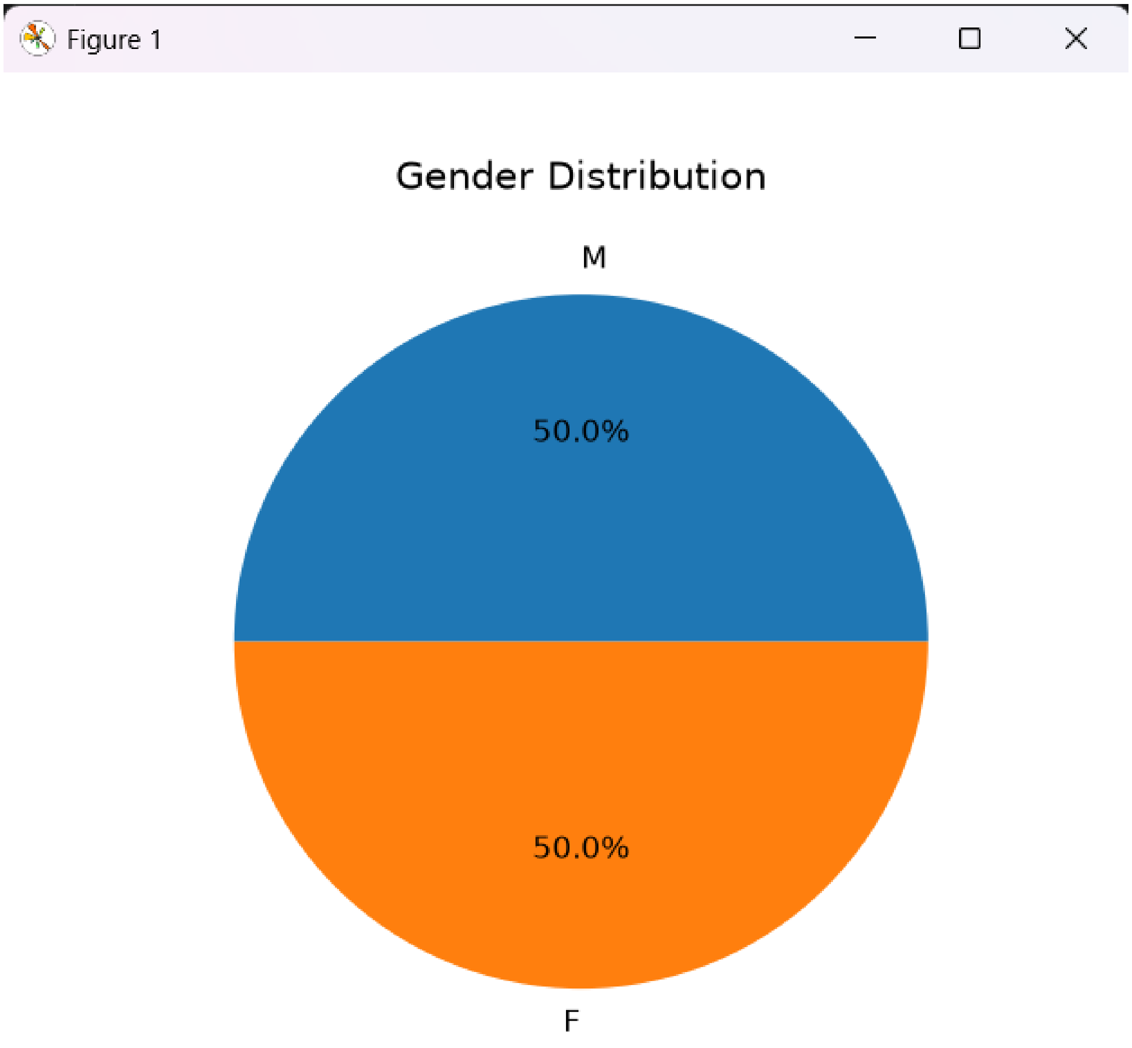
Output:



Piechart Code:

```
gender = df["Gender"].value_counts()


plt.figure(figsize=(5,5))

plt.pie(gender, labels=gender.index, autopct="%1.1f%%")

plt.title("Gender Distribution")

plt.show()
```

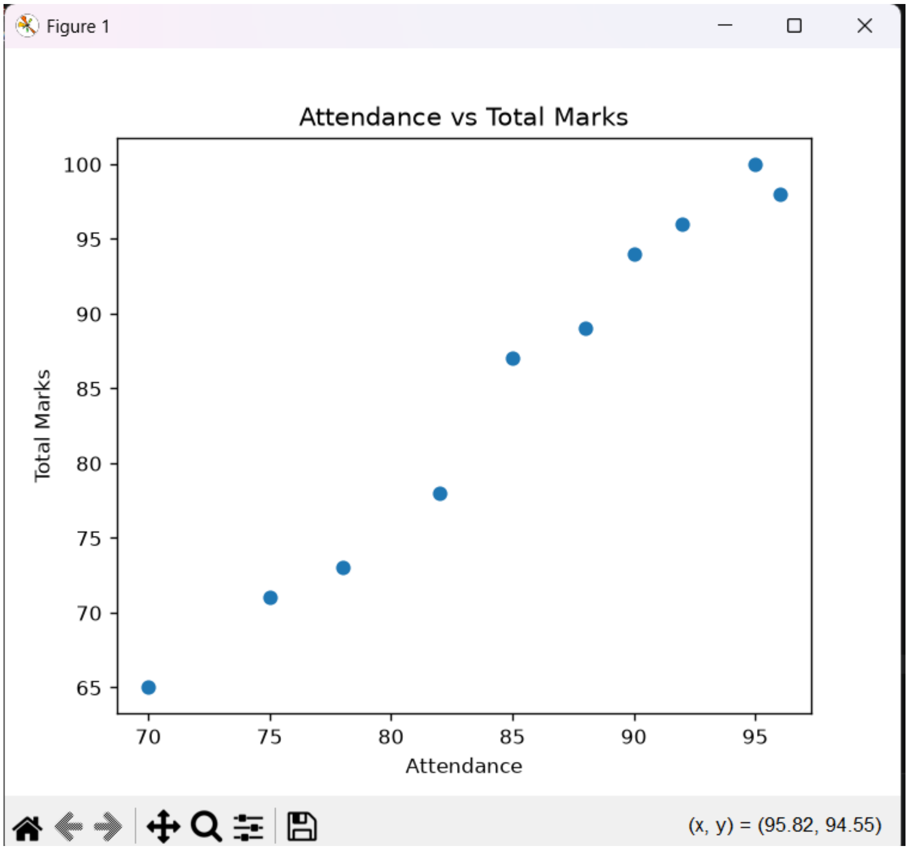
Output:



Scatter plot code:

```
plt.figure(figsize=(6,5))

plt.scatter(df["Attendance"], df["TotalMarks"])

plt.title("Attendance vs Total Marks")

plt.xlabel("Attendance")

plt.ylabel("Total Marks")

plt.show()
```

Output:



Students with higher attendance generally scored higher TotalMarks. This indicates a positive relationship between attendance and academic performance.

Task H: Observations

• Write any three observations based on the analysis performed.

• State whether higher attendance appears to be associated with better academic performance.

Observations:

- Female students have a higher average attendance than male students.
- Anu scored the highest TotalMarks (100).
- Students with attendance above 90% generally achieved higher marks.

Exploratory Data Analysis (EDA) was successfully performed using Pandas and Matplotlib. Statistical analysis, filtering, ranking, grouping, and visualization of the student dataset were completed successfully